

Multiple Choice (10 marks)

1. Which of the following will increase in vitamin D deficiency?
 - (a) The plasma concentration of vitamin B₁₂
 - (b) The plasma concentration of prothrombin
 - (c) The plasma concentration of preprothrombin
 - (d) The plasma activity of acid phosphatase
 - (e) The plasma activity of lipase
2. Which orbital wall is most likely to collapse in a 'blow out' fracture?
 - (a) The medial wall
 - (b) The ethmoidal wall
 - (c) The orbital rim
 - (d) The floor
 - (e) The roof
3. Blood flows from the right ventricle of the heart into which of the following structures?
 - (a) Coronary arteries
 - (b) Inferior vena cava
 - (c) Pulmonary capillaries
 - (d) Right atrium
 - (e) Left atrium
4. The initial length of telomere stand for members of a species is
 - (a) Determined by the individual's lifestyle habits
 - (b) Dependent on the individual's stress levels
 - (c) Randomly determined at birth
 - (d) Dependent on the individual's gender
 - (e) Dependent on the individual's diet and exercise
5. Men are more likely than women to die from
 - (a) Breast Cancer
 - (b) Influenza
 - (c) Stroke
 - (d) Suicide
 - (e) Asthma

True / False (10 marks)

Answer True or False

6. True or False: Engaging in weight lifting alone is sufficient to enhance reaction time in older adults compared to younger adults.
7. True or False: The karyotype 47,XXY is known to have a milder impact on human development compared to 47,XXX.
8. True or False: Newborn infants in countries with limited access to safe water should be exclusively breastfed for six months to ensure optimal nutrition.
9. True or False: DNA polymerase is not classified as a zinc-dependent enzyme in biochemical processes.
10. True or False: The elderly play a crucial role in achieving effective change within a community due to their experience and established networks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of Vitamin E deficiency in dental health, particularly its impact on enamel defects and the increased risk of dental caries. Analyze underlying mechanisms and implications for treatment.
12. Discuss the genetic relationship between siblings and analyze why individuals are most likely to share a common HLA haplotype with their sisters.

End of Paper